

Tanmay Bhatia

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EDUCATION

Johns Hopkins University <i>Master of Science in Mechanical Engineering</i> Relevant coursework: Fluid Dynamics, Fluid Mechanics, Microfluidic Systems, Mechatronics, Product Management	Aug 2024 - May 2026 Baltimore, MD
Nirma University <i>Bachelor of Technology in Mechanical Engineering</i> Relevant coursework: Machine Design, Thermodynamics, Heat and Mass Transfer, Energy Systems, Energy Management	Jul 2019 - May 2023 Ahmedabad, India

SKILLS

Design	CAD, Catia, SolidWorks, Creo, Inventor, AutoCAD, Sheet metal design, DFMA, Electrical Routing, Thermal routing, Circuit design, Plastics design, GD&T, Engineering Drawings, PLM, Windchill
Simulation	Ansys, Fluent, Creo Simulate, COMSOL, CFD, FEA
Manufacturing	Power tools, Procurement Experience, Manufacturing processes, Bill of Materials management
Computer Skills	Python, C, MATLAB, MS Excel, MS Office, MS PowerPoint
Microfabrication	Photolithography, Soft lithography, Plasma bonding, Optical profilometer, PDMS pouring & punching

EXPERIENCE

Graduate Researcher Advisor: Dr. Soojung Claire Hur <i>HUR Laboratory, JHU</i> • Designed microfluidic device layouts in AutoCAD for particle focusing, sorting, and flow analysis • Fabricated microfluidic devices via soft lithography, PDMS processing, and plasma bonding for particle behavior analysis • Developed COMSOL simulations to validate electrode design, placement, and integration for microfluidic devices	Mar 2025 - Nov 2025 Baltimore, MD
Mechanical Engineer <i>Euler Motors Pvt. Ltd.</i> • Designed and routed wire harness assemblies in Creo, producing engineering drawings for production electric trucks • Reduced electronic housing weight by 65% through iterative sheet metal enclosure redesign and optimization in Creo • Achieved 33% copper savings by optimizing electrical and thermal routing across low and high voltage systems • Led low and high voltage electrical architecture design for cross-platform compatibility across electric truck platforms • Managed BoM tracking for 400+ components across engineering, procurement, and vendor teams to reduce delays • Conducted powertrain analysis and motor sizing using drive cycle data to optimize EV truck range and efficiency	Jan 2023 - May 2024 New Delhi, India
Mechanical Engineering Intern <i>Flowchem Engineering</i> • Prepared and reviewed GD&T reports for metal castings, ensuring compliance with internal quality control checks • Collaborated with engineers on centrifugal pump selection, optimizing rotor and materials for client application	May 2022 - Jun 2022 Ahmedabad, India
Mechanical Engineering Intern <i>Burner India</i> • Developed 3D CAD models and engineering drawings for industrial burners, used by vendors and installation teams • Supervised batch assembly line for burner fabrication, ensuring consistent product quality and overall process efficiency • Engineered slot-and-teeth interlocking assembly mechanism, reducing burner assembly time by 20%	May 2021 - Jul 2021 Mehsana, India

PROJECTS

Autonomous Hockey Robot • Fabricated robot chassis via laser cutting and led full mechanical assembly and integration for class competition • Wired and assembled robot electronics including Arduino, IR sensor, IMU, Pixy camera, and Zigbee module • Contributed to PID motor control and ArUco tag-based positioning for autonomous maze navigation and competition	Spring 2025
Aerodynamic Advantage of Active Grille Shutters • Built 2D vehicle CAD model in Creo and ran CFD simulations in ANSYS to evaluate drag reduction • Simulated airflow using k-epsilon turbulence modeling, quantifying drag reduction across varying speeds and flap angle	Fall 2024
Piping System Cost Optimization • Modeled a dual-outlet piping system with varying pressure and flow requirements from a single pump station • Optimized pipe diameter and pump selection to minimize total system cost based on material weight and power draw	Spring 2026

LEADERSHIP EXPERIENCE

Lead Math Tutor Henderson Hopkins School, Johns Hopkins University • Led and mentored a team of 10 tutors, supporting lesson delivery and resolving operational challenges • Provided one-on-one math tutoring to 50+ middle school students, building foundational math skills	Feb 2025 - Nov 2025
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